

Week 3 Project – Password Cracking with John the Ripper

Networkwalks Academy — Cybersecurity Lab Report

Objective

Demonstrate password recovery on password-protected PDF files using John the Ripper (JTR) running inside a Kali NetHunter environment on Android, and validate results using an online PDF hash-cracking utility.

Environment & Tools

Component	Detail
Platform	Kali NetHunter for Termux (Android, ARM64)
Cracking tool	John the Ripper 1.9.0-Jumbo-1
Hash extraction tool	pdf2john.pl (bundled with John the Ripper)
Validation tool	Networkwalks online PDF hash cracker
Wordlist	John's default password.lst

Target File 1: My-Locked-PDF1.pdf

Method: Command-line cracking with John the Ripper.

Steps performed:

1. Copied the target file into the working directory:
`cp "/sdcard/Download/My-Locked-PDF1.pdf" ~/`
2. Extracted the password hash:
`pdf2john ~/My-Locked-PDF1.pdf > pdf1.hash`
3. Ran John the Ripper against the hash:
`john pdf1.hash`
4. Verified the cracked password:
`john --show --format=PDF pdf1.hash`

Hash type: PDF [MD5 SHA2 RC4/AES 32/64], Revision 4, Version 4, 128-bit key

Result: Recovered password:

password1

Target File 2: My-Locked-PDF2.pdf

Method: Hash extracted and cracked using the Networkwalks online PDF hash-cracking tool (networkwalks.com) as a supplementary/validation approach. The PDF was parsed locally in-browser and the extracted hash was run against a built-in password list.

[illegible]

password1

```
nw{networkwalks_persistence_jtr_270521}
```

Both target files used a weak, dictionary-guessable password ('password1'), which was cracked almost instantly using a small default wordlist. This highlights a common real-world weakness: predictable passwords based on common words with simple numeric suffixes are trivial for tools like John the Ripper to defeat, even on modest hardware such as a smartphone running a proot-based Linux environment. Strong, unique, high-entropy passwords (or passphrases) are far more resistant to dictionary and brute-force attacks of this kind.

This exercise demonstrated end-to-end password recovery against encrypted PDF files: setting up John the Ripper in a mobile Kali NetHunter environment, extracting crackable hashes with pdf2john, running a dictionary attack, and validating results with an independent online tool. Both target files were successfully cracked, confirming the recovered password and capturing the associated challenge flag.

— End of Report —